

Five Days at Fremont — Part B

AC101 integrated Tesla case · worked in Lecture 2

*Part B of the week-long AC101 Tesla case — worked in Lecture 2. The full fact-vs-fiction note and company background are in the Part A handout (the complete case is on the course site). In short: Tesla is real and the filed figures are real; **every product-costing scenario, internal figure and decision here is hypothetical**; exhibits are labelled REAL, SIMULATED or ESTIMATE. Every question is worked and discussed in the lecture.*

Part B — Tuesday: the price war and the invisible split

Tuesday opens with a forwarded headline and one line from Reyes: “This is why I’m asking.”

Tesla cuts U.S. prices of Models Y, X, S by \$2,000 (CNBC, 20 April 2024). By that cut, the base Model Y stood at **\$42,990** — roughly **\$23,000** below the cheapest Model Y’s January 2023 sticker (then the \$65,990 Long Range), after a single \$13,000 cut that month and repeated smaller cuts since. Like-for-like, the Long Range itself was down about \$18,000, to \$47,990.

The question every analyst asked is the one on your desk: **if Tesla sells more cars at a lower price, does profit go up or down?** You cannot answer it without knowing how much of Tesla’s cost is fixed and how much is variable — volume spreads fixed costs but adds variable ones. And Tesla, like every company, never publishes “our fixed cost is \$X”. All the outside world sees is total cost and volume, period by period. Recovering the split from those totals is **cost estimation**.

Exhibit B-1 gives you the raw material: eight quarters of real Compustat data, 2023–24 — revenue, COGS, production and deliveries. Before any method, one judgement call: the file has both *production* (cars built) and *deliveries* (cars sold). Which is the right activity driver for COGS? Think about what COGS is before you look at the data; then note that the data agrees with you (Exhibit B-2’s footnote).

The quick-and-dirty method is **high-low**: take the quarter with the highest activity and the one with the lowest, and assume the entire cost difference between them is variable:

$$b \text{ (variable cost per unit)} = (\text{Cost}_{\text{high}} - \text{Cost}_{\text{low}}) \div (\text{Units}_{\text{high}} - \text{Units}_{\text{low}}), \text{ then } a \text{ (fixed cost)} \\ = \text{Cost}_{\text{high}} - b \times \text{Units}_{\text{high}}.$$

It takes two minutes with a calculator — and it listens to only two quarters, throwing the rest away. If either extreme was an unusual quarter, the whole line is wrong, and high-low cannot even tell you how much to trust it.

Which is why the finance team had already fitted the line another way before you arrived. **Ordinary least squares (OLS) regression** draws the single straight line that sits as close as possible to *all* the points at once, and its two pieces are exactly what you need: the **intercept** is the fixed cost, the **slope** the variable cost per car. Their output is Exhibit B-2: an estimated fixed cost of **\$4,295m per quarter**, a variable cost of **\$31,951 per car**, an R^2 of **0.842** and a p-value of **0.0013**. In plain words: the line explains about 84% of the movement in quarterly COGS, and a relationship this tight is very unlikely to be luck — although with eight data points, treat any estimate as indicative rather than gospel. One caveat travels with it: the COGS column is Compustat’s, and Compustat reports COGS with depreciation, amortisation and impairment shown separately rather than inside the line (\$5,368m in FY2024) — so the intercept *understates* Tesla’s total fixed production cost.

Now the split earns its keep. **Cost–volume–profit** logic runs on one defined term: **contribution margin** — selling price minus variable cost per unit. It is *not* profit; it is what each car contributes towards a pile of

fixed cost, and only once the pile is gone does contribution become profit. Break-even is simply the number of contributions it takes to fill the pile: $Q^* = \text{Fixed cost} \div \text{CM}$. For a revenue figure per car, the best the outside world offers is **\$54,599** – 2024 revenue of \$97,690m spread over 1,789,226 deliveries. Handle it with the label on: it folds energy, services and regulatory-credit revenue over the cars, so it sits well above the ~\$43k Model Y sticker. It is a proxy, not an average selling price.

Run the numbers (Question 4) and Tuesday's resolution emerges: against its COGS break-even, Tesla's actual 2024 delivery run-rate carries a **wide margin of safety** – deliveries could fall by more than half before quarterly contribution stopped covering the estimated fixed pile; Question 4 puts a number on it. On production economics, the price cut does not sink the company – *provided the volume holds*. The caveat belongs in the same sentence: this break-even covers the cost of *making* the cars only. SG&A and R&D – some \$9.7bn a year in Exhibit A-1 – still sit on top before real operating profit. Cost estimation has not given you certainty; it has given you a number to argue over, which is what the whole price-war debate was missing.

Assignment questions – Part B

- Exhibit B-1 has both production and deliveries. Which is the right activity driver for COGS, and why?
- High-low on Exhibit B-1 (work in \$m; deliveries are in cars): identify the high and low quarters by deliveries, then compute b and a. Which six data points did you just throw away?
- Compare your high-low line with Exhibit B-2's OLS line (\$4,295m + \$31,951 per car). Why do they differ? Make the case for the line you would take to the CFO, and say what $R^2 = 0.842$ and $p = 0.0013$ each tell you, in plain words.
- CVP with the OLS estimates: compute the contribution per car, the break-even volume per quarter, and the margin of safety against actual average 2024 deliveries. So: does the price cut, by itself, sink Tesla? State the one caveat about what this break-even does and does not cover.
- Sensitivity: suppose revenue per car delivered falls a further 10%, to \$49,139. Recompute the contribution per car and the break-even. Then one paragraph: why might the straight line itself mislead outside the range of volumes you observed – and what does the “sticky costs” evidence (Anderson, Banker & Janakiraman 2003, *Journal of Accounting Research*) say about the symmetric $a + bX$ assumption?
- A colleague ranks four possible drivers for quarterly COGS by how well each fits the eight quarters (Exhibit B-3) and recommends total revenue – “the highest R-squared on the list, 0.963”. Which driver should you use to estimate Tesla's COGS cost function? Give, in one line each, the reason you reject the other three. For the revenue line, your reason must say what its fitted intercept of minus \$573m and its slope of 0.791 actually mean.
- Tesla issues delivery guidance of 500,000 cars for next quarter. Using the estimated cost function in Exhibit B-2, forecast next quarter's COGS, then compute the projected quarterly contribution at the labelled revenue-per-car proxy. Exhibit B-4 lists six planning figures; three of them do not belong in this forecast – identify each and say why in one line. Finally, state the assumption your forecast relies on about the 500,000 figure, and name what still sits on top of contribution before Tesla would reach operating profit.

Exhibits – Part B

Exhibit B-1 – Eight quarters of real data, 2023–24 REAL – Compustat, quarterly, 2023–24. Note: Compustat reports COGS with depreciation, amortisation and impairment shown separately (\$5,368m in FY2024); a fixed-cost estimate built on this series therefore understates total fixed production cost.

Quarter	Revenue (\$m)	COGS (\$m)	Production (cars)	Deliveries (cars)
Q1 2023	23,329	17,772	440,808	422,875
Q2 2023	24,927	19,240	479,700	466,140
Q3 2023	23,350	17,937	430,488	435,059
Q4 2023	25,167	19,497	494,989	484,507
Q1 2024	21,301	16,359	433,371	386,810
Q2 2024	25,500	19,644	410,831	443,956
Q3 2024	25,182	18,837	469,796	462,890

Quarter	Revenue (\$m)	COGS (\$m)	Production (cars)	Deliveries (cars)
Q4 2024	25,707	20,032	459,445	495,570

Exhibit B-2 – The finance team’s regression output *ESTIMATE* computed from the REAL Compustat data in Exhibit B-1 (OLS, quarterly COGS in \$m against deliveries, 8 quarters). An estimate, not a figure Tesla discloses.

Output	Value
Intercept (fixed cost)	\$4,295m per quarter
Slope (variable cost)	\$31,951 per car
R ²	0.842
p-value (slope)	0.0013

Footnote: regressed on production instead of deliveries, R² collapses to 0.147.

*CVP input: 2024 revenue per car delivered = \$97,690m ÷ 1,789,226 = **\$54,599** – revenue per car delivered includes energy, services and regulatory credits, so it sits above the ~\$43k Model Y sticker; a proxy, not an ASP.*

Exhibit B-3 – A colleague’s shortlist of drivers for quarterly COGS ESTIMATE – R², fitted intercept and slope computed from the REAL Exhibit B-1 data (8 quarters, OLS, quarterly COGS in \$m); the Supercharger row is *SIMULATED* (illustrative).

Proposed driver	R ²	Fitted intercept	Fitted slope	Note
Deliveries (cars sold)	0.842	\$4,295m	\$31,951 / car	the Exhibit B-2 line
Production (cars built)	0.147	\$11,153m	\$16,603 / car	–
Total revenue (\$m)	0.963	–\$573m	0.791 per \$ of revenue	highest R ² on the list
Supercharger stalls energised	–	–	–	series recorded only from Q3 2023 (6 of the 8 quarters)

Exhibit B-4 – Next-quarter planning sheet (for the COGS forecast) *SIMULATED* – illustrative planning figures; each \$ magnitude is set near Tesla’s real FY2024 quarterly run-rate (10-K SG&A \$5,150m/yr; regulatory-credit revenue ~\$2.76bn/yr) and labelled – not Tesla’s actual plan.

Item	Figure
Delivery guidance, next quarter	500,000 cars
Estimated cost function (from Exhibit B-2)	COGS (\$m) = 4,295 + 0.031951 × cars
Revenue per car delivered (proxy, from Part B)	\$54,599
Allocated corporate SG&A, next quarter	\$1,290m
Battery-line retooling, paid last quarter	\$900m
Regulatory-credit revenue, next quarter	\$650m